

A Geometric-Electronic Principle for Curvature-Driven Single-Atom Catalysis

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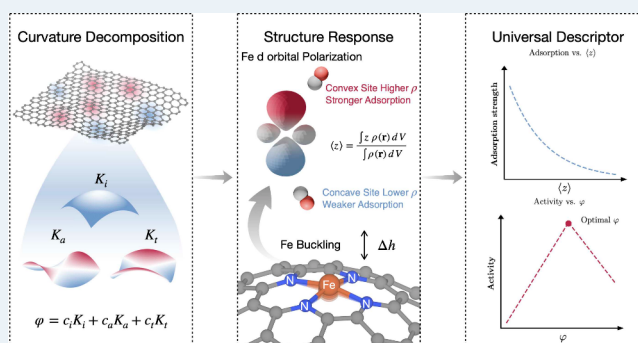
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ABSTRACT: Substrate curvature offers a powerful handle for catalytic tuning, yet a general quantitative framework for arbitrary anisotropic morphologies remains lacking. Here, by combining differential geometry with a symmetry-constrained Taylor expansion, we derive a geometric descriptor, φ , from the full curvature tensor that compresses complex local curvature into a predictive variable for catalytic properties. Using Fe–N₄–C single-atom catalysts as a model for CO₂RR, systematic calculations across 602 nonequivalent curved sites reveal that the curvature-dependent activity variation is governed by the out-of-plane *d*-orbital centroid, $\langle z \rangle$. This electronic descriptor captures the asymmetric polarization of *d* orbitals along the surface normal and naturally distinguishes convex from concave active sites. We trace this electronic polarization to its mechanical origin: the intrinsic stiffness contrast between pentagonal and hexagonal rings converts geometric strain into opposite vertical displacements of the metal center that drive the orbital response. This establishes a quantitative causal pathway from geometry (φ) through mechanics and electronic structure ($\langle z \rangle$) to catalytic activity. This framework elevates local curvature from a qualitative structural feature to a quantitative and tunable design axis for catalyst engineering.

KEYWORDS: CO₂ reduction reaction, curvature effect, descriptors, first-principles calculations, single-atom catalysis



1. INTRODUCTION

In the pursuit of tailored functionality, surface curvature has emerged as a fundamental yet underutilized physical dimension,^{1,2} offering a design avenue strictly orthogonal to conventional chemical composition tuning.^{3–8} This geometric modulation constitutes a universal principle that transcends material classes, spanning from the pre-organized, concave pockets of metalloenzymes^{9,10} to the continuously deformable topologies of two-dimensional materials, which can host spherical,¹¹ cylindrical,¹² and saddle-shaped geometries.^{13–15} In this landscape, harnessing substrate curvature represents a transformative strategy to precision-engineer single-atom catalysts (SACs). Given that SACs are predominantly anchored on flexible 2D scaffolds with active sites strictly governed by their local coordination environment,^{16–18} curvature serves as a potent lever to modulate lattice strain^{19,20} and fine-tune their electronic properties.²¹

Despite this profound impact, establishing a quantitative, predictive principle that links this tensor geometry to catalytic performance remains an elusive challenge. Experimentally, realistic supports exhibit morphological nonuniform landscapes with broad curvature distributions.^{22–30} Catalytic measurements inevitably yield convoluted ensemble averages, making it nearly impossible to deconstruct and isolate the specific contributions

of distinct geometric features. Theoretically, current interpretations are largely based on idealized, uniaxial bending models (e.g., carbon nanotubes (CNTs)^{31–42}) or isotropic models (e.g., spherical carbon^{28,29}). While these approaches capture a linear dependence of binding strength on curvature magnitude, they fundamentally overlook the complex biaxial and anisotropic topologies inherent to realistic surfaces.²⁸ In addition, the vectorial nature of the electronic response induced by curvature (e.g., the inequivalence of convex and concave sides) often confounds conventional scalar descriptors (e.g., the *d*-band center), which are incapable of resolving such directionality. Even for convex surfaces, reported findings are inconsistent: while some studies report that increasing curvature strengthens adsorption,⁴⁰ others observe the exact opposite trend^{37–39} under seemingly similar conditions. This inconsistency strongly suggests that interpretations relying solely on simplified geo-

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metric models (i.e., scalar curvature magnitude) are insufficient to capture the intrinsic complexity of real and anisotropic environments. Collectively, these challenges call for a universal framework that quantitatively links local tensor geometry to catalytic activity.

Herein, we establish a unified geometric–electronic framework that addresses these challenges by defining a structure–electronics–activity causal chain (Figure 1). Grounded in differential geometry, the local curvature at the active site in M–N–C is parameterized by a quadratic paraboloid defined by principal curvatures (k_1 , k_2) and orientation (θ). The curvature tensor is then decomposed into three orthogonal modes—isotropic (K_i), anisotropic (K_a), and torsional (K_t). For the M–N₄ motif, point-group selection rules remove odd-order K_t terms; under the small-curvature linear approximation, this reduces to $\varphi = c_i K_i + c_a K_a$. Using Fe–N₄–C as a model system with 602 nonequivalent curved sites, we show that an intrinsic stiffness contrast between pentagonal and hexagonal rings converts geometric strain into opposite vertical displacements of the metal center. This mechanical response induces asymmetric out-of-plane *d*-orbital polarization, captured by the *z*-centroid descriptor ($\langle z \rangle$). It links geometry (φ), mechanics, electronics ($\langle z \rangle$), and activity (ΔG_{max}), producing a predictive volcano relation for the CO₂ reduction reaction (CO₂RR). The framework is further validated across metal centers, adsorbates, coordination environments, and periodic models and remains robust under applied electrochemical-potential conditions, supporting local curvature as a quantitative catalyst-design axis.

2. COMPUTATIONAL DETAILS

All spin-polarized DFT calculations were performed using the Vienna Ab initio Simulation Package (VASP).^{43,44} The projector

augmented wave (PAW) method⁴⁵ was employed to describe the ion–electron interactions. To better describe the adsorption properties on the surface, the revised Perdew–Burke–Ernzerhof⁴⁶ (RPBE) generalized gradient approximation⁴⁷ (GGA) functional was used for the exchange–correlation energy. The van der Waals interactions were accounted for using the DFT–D3 correction by Grimme.^{48,49} A plane-wave cutoff energy of 500 eV was used. All structural models were placed in a supercell with dimensions of 24 × 22 × 16 Å to minimize interactions between periodic images. Due to the large size of the supercell, the Brillouin zone was sampled using a Γ -centered 1 × 1 × 1 *k*-point mesh. The convergence criteria for electronic self-consistency and ionic relaxation were set to 10^{−5} eV and 0.02 eV/Å, respectively. In all calculations, a Hubbard–U parameter⁵⁰ of $U_{\text{eff}} = 2.0$ eV was applied to the Fe 3*d* orbitals, consistent with previous benchmark studies.^{51,52}

To take into account the effects of solvation and applied potentials on the reaction free energies, we employed the non-linear, nonlocal implicit solvation model as implemented in the VASPsol++ package.⁵³ An aqueous electrolyte with a concentration of 1.0 M was modeled using an effective ionic radius of 4.0 Å. Constant potential calculations were performed to model the system under an applied bias within the grand canonical ensemble. Based on recent benchmarks for the RPBE functional using nonlocal solvation models,⁵⁴ the absolute potential of the Standard Hydrogen Electrode (SHE) was taken as −4.44 eV relative to the vacuum level. Accordingly, to simulate an applied potential of $U = -0.8$ V vs. SHE, the Fermi level reference parameter was set to −3.64 eV. The Gibbs free energy change (ΔG) for each elementary step of CO₂RR was computed as follows:

$$\Delta G = \Delta E + \Delta E_{\text{ZPE}} - T\Delta S \quad (1)$$

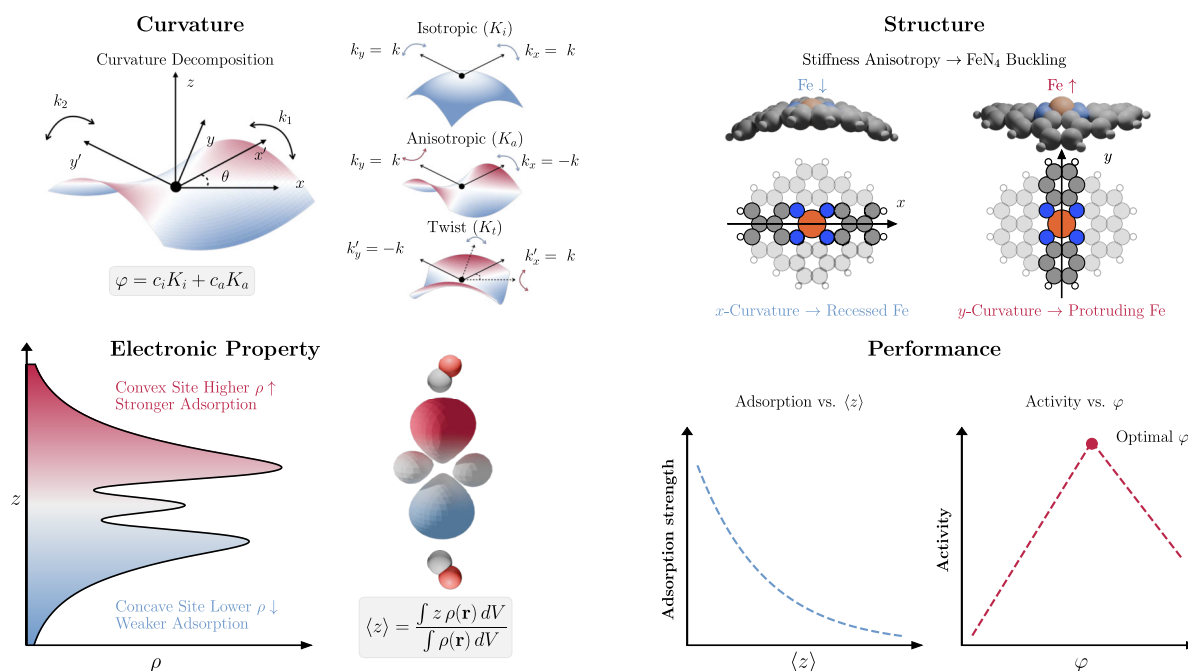


Figure 1. Curvature–structure–electronic coupling governing catalytic behavior. Local surface curvature, decomposed into isotropic (K_i), anisotropic (K_a), and twisting (K_t) modes, defines the geometric descriptor φ . The intrinsic stiffness contrast translates geometric curvature into directional buckling of the Fe–N₄ center, generating the asymmetric out-of-plane *d*-orbital polarization. This electronic response directly modulates adsorption energetics, establishing a volcano relationship between φ and catalytic performance.

where ΔE is the difference in the grand canonical potential between the adsorbed state and the reference state at a fixed potential. The zero-point energy correction (ΔE_{ZPE}) and entropy contributions ($T\Delta S$) were obtained from vibrational frequency calculations. To account for the systematic errors of the RPBE functional in describing the gas-phase CO_2 molecule, a correction of +0.45 eV was applied to its total energy, following the standard correction scheme established by Nørskov and co-workers.^{55,56} Crystal Orbital Hamilton Population (COHP) and integrated COHP (ICOHP) analyses were performed using the LOBSTER package^{57,58} to quantify the intrinsic bond strengths between the first and second coordination shells along the pentagonal and hexagonal ring directions. Post-processing of the calculated data was facilitated by the VASPKIT code.⁵⁹

3. RESULTS AND DISCUSSION

3.1. Geometric Descriptor from Curvature Tensor and Symmetry

To rigorously quantify the geometric influence, we leverage the formalism of differential geometry to construct a unified theoretical framework (see Note S1 for complete derivation). The local topology of any smooth curved surface can be approximated by a quadratic paraboloid:

$$z(x, y) = -\frac{1}{2} [k_1 (x \cos \theta + y \sin \theta)^2 + k_2 (-x \sin \theta + y \cos \theta)^2] \quad (2)$$

This equation defines the surface height $z(x, y)$ relative to the local tangent plane, where k_1 and k_2 are the principal curvatures, and θ denotes the orientation of their principal directions. From these principal curvatures, the Gaussian curvature is defined as their product, $K_G = k_1 k_2$. This intrinsic metric provides a rigorous criterion for classifying the local topological nature of the catalyst surface. As schematically illustrated in Figure S1, the sign of K_G distinguishes three fundamental geometric motifs: (a) elliptic regions ($K_G > 0$), where the surface curves in the same direction (synclastic), forming dome- or bowl-shaped sites; (b) parabolic or planar regions ($K_G = 0$), which correspond to cylindrical surfaces or flat facets; and (c) hyperbolic regions ($K_G < 0$), characterized by opposing principal curvatures (anticlastic) that create saddle-shaped geometries.

While (k_1, k_2, θ) fully parameterize the local curvature, they are inherently coupled. Expanding eq 2 yields the curvature tensor components in the (x, y) frame (k_x, k_y, k_{xy}) , which can be further reorganized into a set of orthogonal angular modes via polar decomposition:

$$K_i = k_x + k_y = k_1 + k_2 \quad (3)$$

$$K_a = k_x - k_y = (k_1 - k_2) \cos 2\theta \quad (4)$$

$$K_t = 2k_{xy} = (k_1 - k_2) \sin 2\theta \quad (5)$$

Here, k_x and k_y quantify the geometric bending along the axes aligned with the x - and y -axes, respectively. K_i represents the isotropic mean curvature, quantifying the overall magnitude of surface bending. Physically, this corresponds to a uniform breathing mode that induces average structure bending without breaking in-plane symmetry. In contrast to the isotropic term K_i , both K_a and K_t represent symmetry-breaking anisotropic

distortions that contribute zero to the mean curvature. However, they capture distinct physical modes: K_a quantifies the directional disparity aligned with the intrinsic axes (x and y), representing a normal shear deformation. The mixed term K_t represents twist curvature, capturing the off-axis shear distortion that twists the local coordination environment.

Under this framework, any intrinsic property $\mathcal{P}$ of an active site embedded in such a surface is functionally dependent on (K_i, K_a, K_t) . A multivariate Taylor expansion around the planar limit ($K_i = K_a = K_t = 0$) gives the general curvature–property relationship:

$$\mathcal{P}(K_i, K_a, K_t) = \mathcal{P}_0 + \sum_{j \in \{i, a, t\}} c_j K_j + \frac{1}{2} \sum_{j, l \in \{i, a, t\}} H_{jl} K_j K_l + \mathcal{O}(K^3) \quad (6)$$

where $c_j = (\partial \mathcal{P} / \partial K_j)_0$ are the linear geometric susceptibilities and $H_{jl} = (\partial^2 \mathcal{P} / \partial K_j \partial K_l)_0$ are the elements of the geometric Hessian, with $j, l \in \{i, a, t\}$. This expansion is model-agnostic: the geometric structure (orthogonal modes and their coupling pattern) is universally dictated by differential geometry, while all material-specific information is encoded in the response coefficients (c_j, H_{jl} , etc.).

The functional form of eq 6 is further constrained by the point-group symmetry of the active site. For the $\text{M}-\text{N}_4$ moiety (D_{2h}), the twist mode K_t is antisymmetric under vertical mirror reflections, which requires all odd-order terms in K_t —including the linear coefficient c_t and all mixed terms of the form K_t^{2n+1} —to vanish identically (Note S1.4). This is an exact selection rule, independent of curvature magnitude. The leading contribution of K_t therefore enters only at second order ($\propto K_t^2$). Furthermore, in the small-curvature limit, where the second-order and higher terms become negligible, this further simplifies to the linear geometric descriptor:

$$\varphi = c_i K_i + c_a K_a \quad (7)$$

Thus, the combination of symmetry selection and linearization reduces the general expansion (eq 6) from a multi-term tensor series to a two-parameter scalar. The framework is valid for continuous geometric modulations where the local curvature varies smoothly; it is not intended to describe abrupt topological defects or structural singularities at which differentiability breaks down.

3.2. Construction of the Curvature-Defined Structural Library

Applying this formalism to the $\text{Fe}-\text{N}_4-\text{C}$ system as a primary case study, we constructed a systematic library of curved structures by geometrically projecting a planar, hydrogen-passivated graphene sheet embedding a $\text{Fe}-\text{N}_4$ moiety onto analytically defined quadratic surfaces (Figure S2). This projection procedure enables controlled and independent variation of the three geometric parameters (k_1, k_2, θ) , thereby providing a well-defined curvature space for systematic exploration. Previous studies have demonstrated that cluster models reproduce qualitative trends and dopant-dependent reactivity patterns in SACs.^{60–62} Since our objective is to establish curvature–property scaling based on relative energetic modulation across geometries, this modeling strategy is appropriate. In the defined coordinate system, the x - and y -axes symmetrically

bisect the six- and five-membered rings, respectively (Figure 1). During structural relaxation, the carbon scaffold was constrained to preserve the imposed curvature field (Figure S3a), while allowing the Fe–N₄ moiety and adsorbates to fully relax (details in Note S2).

To establish the physically relevant curvature regime, we analyzed the surface topology of amorphous carbon generated from molecular dynamics simulations, with curvature statistics evaluated on the *sp*² framework (see Note S3). The simulated support is predominantly composed of *sp*²-hybridized carbon (92.9%), and root-mean-square deviation analysis confirms that the graphitic scaffold remains dynamically stable throughout the trajectory, with fluctuations of merely 0.22 Å (Figure S4a–d). The mean curvature ($K_i/2$) displays a narrow, symmetric distribution centered near zero, with 81.9% of the surface area confined within ± 0.12 Å^{−1} (i.e., $|K_i| \leq 0.24$ Å^{−1}; Figure S4e). This range is fully consistent with experimentally reported local curvatures of 0.00–0.09 Å^{−1} in porous carbon supports,²⁸ confirming that our parameter space reflects realistic structural fluctuations. Within this experimentally validated regime, Gaussian-curvature analysis reveals that 76.7% of the active *sp*² surface exhibits negative Gaussian curvature ($K_G < 0$), demonstrating that saddle-shaped, anisotropic geometries are the dominant topological motif in amorphous carbon frameworks (Figure S4f). Notably, such biaxial saddle configurations are intrinsically absent from prevailing theoretical models that approximate curvature using one-dimensional (CNTs, $K_G = 0$) or strictly isotropic (spherical fullerenes, $K_G > 0$) constructs, thereby omitting the majority curvature state realized in practical supports.

Within this physically relevant regime, k_1 and k_2 were systematically sampled in increments of 0.02 Å^{−1} (Figure S5a), and θ was varied from 0 to $\pi/2$ (the irreducible domain defined by Fe–N₄ symmetry) in steps of $\pi/12$ (Figure S5b), initially defining 1,183 curved geometries. In planar Fe–N₄–C, the upper and lower surfaces are equivalent due to horizontal mirror symmetry (σ_h); however, curvature breaks this symmetry, creating electronically distinct convex and concave faces. While this ostensibly doubles the adsorption configurations to 2366, the geometric symmetry of the quadratic form dictates that a configuration on the lower surface (k_1, k_2, θ , bottom) is physically mapped to its mirror image on the upper surface ($-k_1, -k_2, \theta$, top). Accordingly, all adsorbates in this work were positioned above the Fe site on the upper surface. After exploiting additional geometric equivalences (principal-axis permutation and spherical isotropy; Figure S6 and Note S4), the dataset was rigorously reduced to 602 unique adsorption sites, forming the complete library used in this study.

3.3. Curvature-Driven Activity Landscape

To quantify how local curvature influences catalytic performance, we computed the complete Gibbs free-energy profiles for the CO₂RR on 602 nonequivalent Fe–N₄–C sites at a potential of −0.8 V vs. SHE, a typical experimental condition, using VASPsol+⁵³ within the constant-potential method. The reaction proceeds via the initial adsorption of CO₂, followed by sequential hydrogenation to form *COOH and *CO intermediates, and finally the desorption of product CO (adsorption configurations illustrated in Figure 2a):



The rate-determining step, characterized by the maximum Gibbs free-energy change (ΔG_{max}), was employed as the overall activity descriptor. This thermodynamic framework is well established for CO₂-to-CO electrocatalysis on M–N₄ sites, including the potential considered here (−0.8 V vs. SHE), where key elementary barriers are often reasonably approximated by the corresponding reaction free energies.^{51,52} To explicitly verify this approximation within our framework, we performed CI-NEB calculations in vacuum for the *COOH → *CO + H₂O step on Fe–N₄–C across the curvature range; the activation barriers decrease monotonically with φ ($R^2 = 0.67$; Figure S7), confirming that the kinetic and thermodynamic descriptors are co-monotonic, in line with a Brønsted–Evans–Polanyi-type relation.

Systematic variation of the principal curvatures (k_1, k_2) and orientation angle (θ) reveals that local curvature enables continuous tuning of the adsorption free energies of *CO₂, *COOH, and *CO over broad ranges of 0.58, 0.74, and 0.76 eV, respectively (Figure 2b). Crucially, linear scaling relationships persist across all sites (Figures 2c,d and S8). Notably, the correlation between *CO₂ and *CO ($R^2 = 0.86$) is stronger than that between *COOH and *CO ($R^2 = 0.67$), hinting at a differential sensitivity of these intermediates to local curvature. The symmetric adsorption geometry of *CO renders it the most robust representative metric for this analysis. Consequently, plotting the reaction free energies of the two potential rate-determining steps (i.e., CO₂ activation and CO desorption) against $\Delta G_{*\text{CO}}$ reveals a clear Sabatier volcano relationship (Figure 2e). The catalytic activity is governed by ΔG_{max} , which follows the lower envelope of these competing trends, reflecting the fundamental trade-off between the two processes. The other two potential-dependent protonation steps remain thermodynamically favorable at −0.8 V vs. SHE (Figure S9).

As anticipated, k_1, k_2 , and θ are intrinsically coupled. Projecting this orderly chemical response back onto the raw geometric coordinates reveals a chaotic landscape. As shown in Figure 3a, the *CO adsorption energy landscape defined by (k_1, k_2, θ) appears highly convoluted. Quantitative analysis via quadratic fitting confirms this intricacy (Figures 3b, S10 and Table S1), revealing erratic gradients and non-monotonic orientation dependencies. This demonstrates that the raw curvature parameters (k_1, k_2, θ) fail to serve as intuitive physical descriptors.

To disentangle this landscape, we transition from the raw coordinates to the decoupled orthogonal basis (eqs. 3–5). By imposing symmetry-constrained sampling conditions to isolate individual curvature components, we obtain well-defined and robust energetic responses for the Fe–N₄ site. When sampling along the isotropic (K_i) and anisotropic (K_a) curvature modes with the remaining components fixed to zero, both modes exhibit clear parabolic dependencies fitted to the symmetry-constrained data points, with $R^2 = 0.93$ for K_i and 0.98 for K_a (Figure 3c,d). Despite their similar functional form, the two curvature modes lead to markedly different response amplitudes. Anisotropic curvature (K_a), which breaks the in-plane

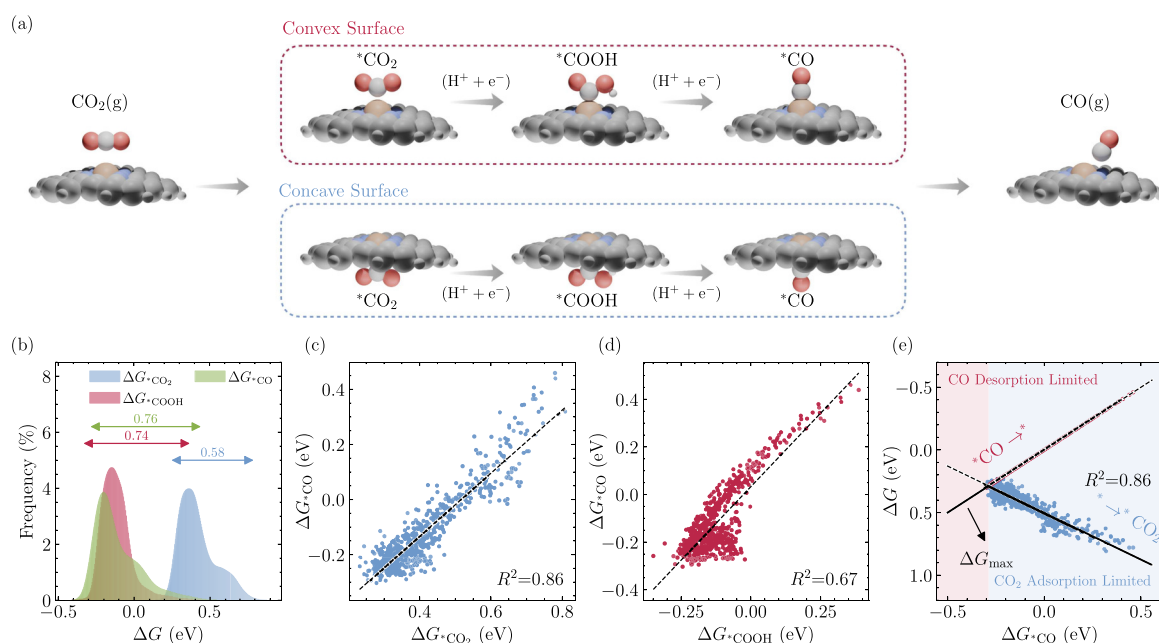


Figure 2. (a) Optimized adsorption configurations of the key reaction intermediates (*CO₂, *COOH, and *CO) on the curved Fe-N₄-C. (b) Probability density distributions of the adsorption free energies for *CO₂, *COOH, and *CO across all computed active sites. Linear scaling relationships between the adsorption free energies of (c) the *CO₂ and *CO intermediates; (d) *COOH and *CO intermediates. (e) Sabatier volcano plot correlating the maximum Gibbs free-energy change (ΔG_{max}) with ΔG_{*CO} as the activity descriptor. The activity is governed by the lower envelope of the scaling relations for CO₂ activation and CO desorption.

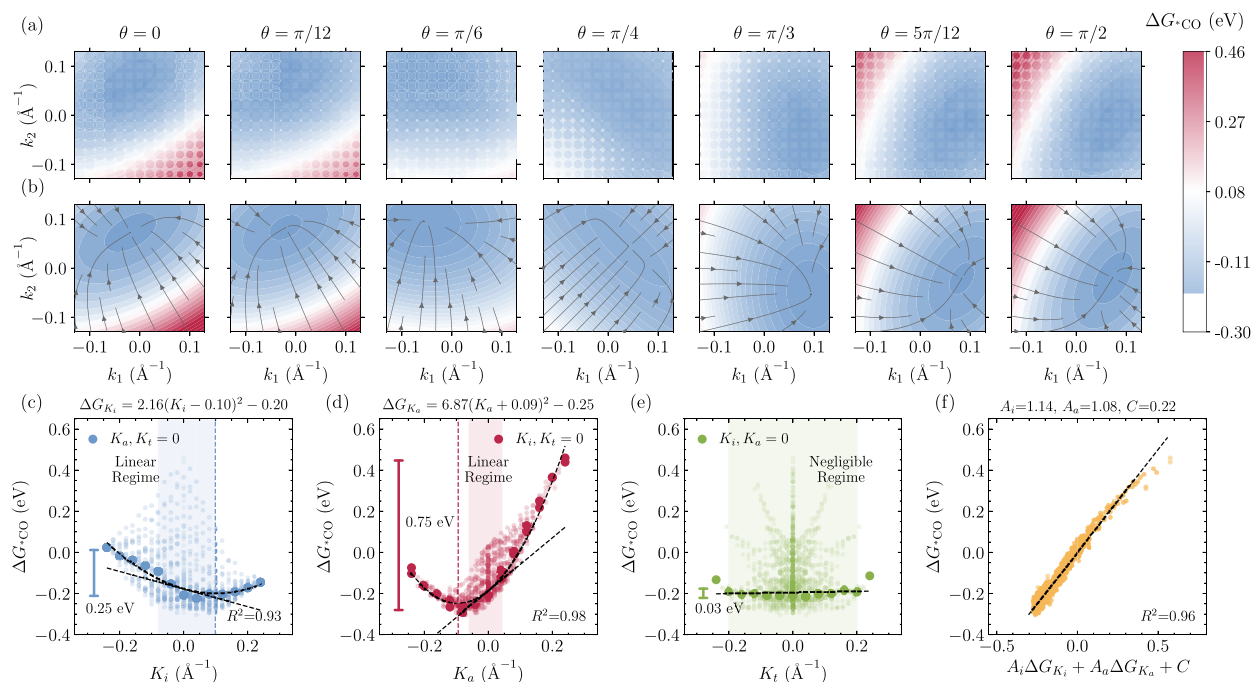


Figure 3. (a) Calculated *CO adsorption free energy (ΔG_{*CO}) as a function of the principal curvatures (k_1 , k_2) at a fixed orientation angle θ . (b) Heatmap of the ΔG_{*CO} landscape from (a) fitted by a quadratic function of k_1 and k_2 . The arrows represent the gradients derived from this fitted surface, indicating the direction of strongest adsorption. (c) Correlation between ΔG_{*CO} and the isotropic curvature parameter K_i , (d) anisotropic curvature parameter K_a , and (e) twist curvature parameter K_t . In panels (c–e), transparent markers correspond to all sampled configurations, whereas large filled markers indicate the points obtained from controlled symmetry-constrained sampling. (f) Parity plot comparing DFT-calculated ΔG_{*CO} with values predicted by the multivariate linear regression model based on ΔG_{K_i} and ΔG_{K_a} .

symmetry of the Fe–N₄ moiety, strongly modulates ΔG_{*CO} (up to ~ 0.75 eV), whereas the isotropic mode (K_i) induces a much weaker change (only ~ 0.25 eV); correspondingly, the quadratic curvature response is substantially larger for K_a than for K_i (quadratic coefficients: 6.87 vs. 2.16). Meanwhile, the torsional curvature component K_t shows no statistically significant correlation with the adsorption energy (Figure 3e). Within the physically relevant range of -0.20 to 0.20 Å⁻¹, the associated energy variation remains negligible (~ 0.03 eV). The vanishing of the linear response is mandated by the D_{2h} symmetry of the Fe–N₄ site, but the near-complete suppression of even the symmetry-allowed second-order response ($\propto K_i^2$) is striking—particularly given that K_a and K_t differ only by a $\pi/4$ rotation of the bending axis yet produce responses differing by more than an order of magnitude. This contrast suggests that the directional bonding topology of the coordination environment strongly dictates the curvature response, rendering purely geometric descriptions insufficient. Furthermore, the absence of mirror symmetry between opposite signs of K_i and K_a exposes the breakdown of descriptions that implicitly assume convex and concave adsorption to be related by simple sign reversal, thereby limiting the applicability of scalar descriptors such as the d -band center. Interestingly, beyond the well-defined individual response laws, the adsorption energy exhibits an almost additive response to the two curvature modes. The adsorption energy can be accurately reconstructed as a linear superposition of the two single-mode responses:

$$\Delta G_{*CO} \approx A_i \Delta G_{K_i} + A_a \Delta G_{K_a} + C \quad (12)$$

As shown in Figure 3f, this simple reconstruction already captures all 602 data points with high fidelity ($R^2 = 0.96$). The fitted coefficients, $A_i = 1.14$ and $A_a = 1.08$, remain remarkably close to unity, indicating that each curvature mode contributes to the adsorption energetics almost independently. Notably, such additivity is not symmetry-enforced—the D_{2h} point group formally allows $K_i K_a$ cross-coupling—suggesting that the electronic response effectively factorizes along the two geometric channels within the explored regime.

Collectively, the orthogonal decomposition resolves the complex (k_1, k_2, θ) landscape into well-defined mode responses, revealing a clear hierarchy ($K_a > K_i \gg K_t$) and an unexpected additivity between the two active modes. While this reconstruction captures the global curvature dependence with high fidelity, ΔG_{*CO} is a composite thermodynamic observable that convolutes geometric relaxation, charge redistribution, and bond reorganization, and thus does not by itself expose the microscopic origin of curvature sensitivity. To rigorously validate our geometric formalism and transcend this phenomenological description, we must bypass this complex energetic convolution and identify a fundamental intrinsic structural–electronic response.

3.4. Electronic Origin: d -Orbital Polarization

We begin by examining whether the curvature response can be rationalized by conventional energy-level-based mechanisms. The Fe center in the planar Fe–N₄ motif resides in an approximate D_{2h} ligand field, which already lifts the degeneracy of the d manifold almost completely, precluding a conventional Jahn–Teller-type distortion as the driving force. Consistent with this picture, PDOS analysis at representative curvatures reveals only marginal shifts in d -orbital energies relative to the planar

reference (Figure S11), indicating that, within the curvature regime explored here, bending does not induce a significant degeneracy-lifting or spin-state transition. This observation is further corroborated by the weak correlation between $\Delta G_{\max}$ and the d -orbital energy centroid, ε_d (Figure S12), defined here as the occupation-weighted average energy of the Fe $3d$ -projected states. Beyond this weak predictive power, ε_d is an inherently isotropic metric that averages over all orbital orientations and therefore fails to distinguish between the electronically distinct convex and concave faces of a curved active site (Figure 4a). Taken together, these results suggest that, at least for the present system, curvature influences reactivity through effects beyond a uniform d -orbital energy shift, underscoring the need for a descriptor that explicitly incorporates the spatial characteristics of the active site.

To decode how geometric distortion translates into binding strength, we moved beyond scalar energetic metrics to interrogate the spatial topology of the Fe frontier wavefunctions. Our analysis begins by identifying the frontier orbitals that govern the Fe–CO interaction. Projected density of states (PDOS) analysis shows that the adsorbate mainly interacts with the out-of-plane orbitals (d_{xz} , d_{yz} , and d_z^2), while the in-plane components (d_{xy} and $d_{x^2-y^2}$) contribute negligibly (Figure S13e). Among them, the d_z^2 orbital exhibits the strongest hybridization with CO, consistent with its direct alignment along the surface normal. We therefore focused on the spin-down d_z^2 state to probe how curvature modulates its spatial distribution. Principal component analysis (PCA) of the d_z^2 charge density across all curved configurations shows that the first principal component (PC1) captures over 89.4% of the total variance, indicating a single dominant mode of curvature-driven electronic reorganization (details in Note S5). Visualization of the PC1 loading vector reveals that curvature induces an asymmetric reweighting of the d_z^2 orbital lobes along the surface normal (Figure 4c), with opposite lobes gaining or losing density depending on curvature. This observation motivated the construction of a new electronic descriptor, $\langle z \rangle_{d_z^2}$, which quantifies the centroid of the orbital charge density along the z -axis:

$$\langle z \rangle_{d_z^2} = \frac{\int z \rho_{d_z^2}(\mathbf{r}) dV}{\int \rho_{d_z^2}(\mathbf{r}) dV} \quad (13)$$

The $\langle z \rangle_{d_z^2}$ values show a perfect linear correlation with the PC1 scores ($R^2 = 1.00$), validating that $\langle z \rangle_{d_z^2}$ faithfully represents the physical meaning of the dominant PCA mode. Furthermore, the polarization of the d_{xz} , d_{yz} , and d_z^2 orbitals is highly synergistic (Figure S14), allowing us to adopt $\langle z \rangle_{d_z^2}$ (hereafter $\langle z \rangle$) as a robust proxy for the collective response of the out-of-plane frontier orbitals. Importantly, $\langle z \rangle$ possesses an intrinsic antisymmetry with respect to the atomic plane, assigning equal and opposite values to the convex and concave sides of a single active site ($\langle z \rangle_{\text{convex}} = -\langle z \rangle_{\text{concave}}$), enabling it to correctly distinguish the electronically inequivalent convex and concave surfaces. This distinction carries direct chemical implications: a positive $\langle z \rangle$ (typical of convex sites) indicates electron density accumulation toward the adsorbate, which enhances orbital overlap and strengthens binding; conversely, a negative $\langle z \rangle$ (concave sites) signifies density depletion at the interface, weakening the interaction (Figure 4d). Governed by this mechanism, the adsorption energies of $*CO$ exhibit an exponential dependence on $\langle z \rangle$ (Figure 4e). This functional form is consistent

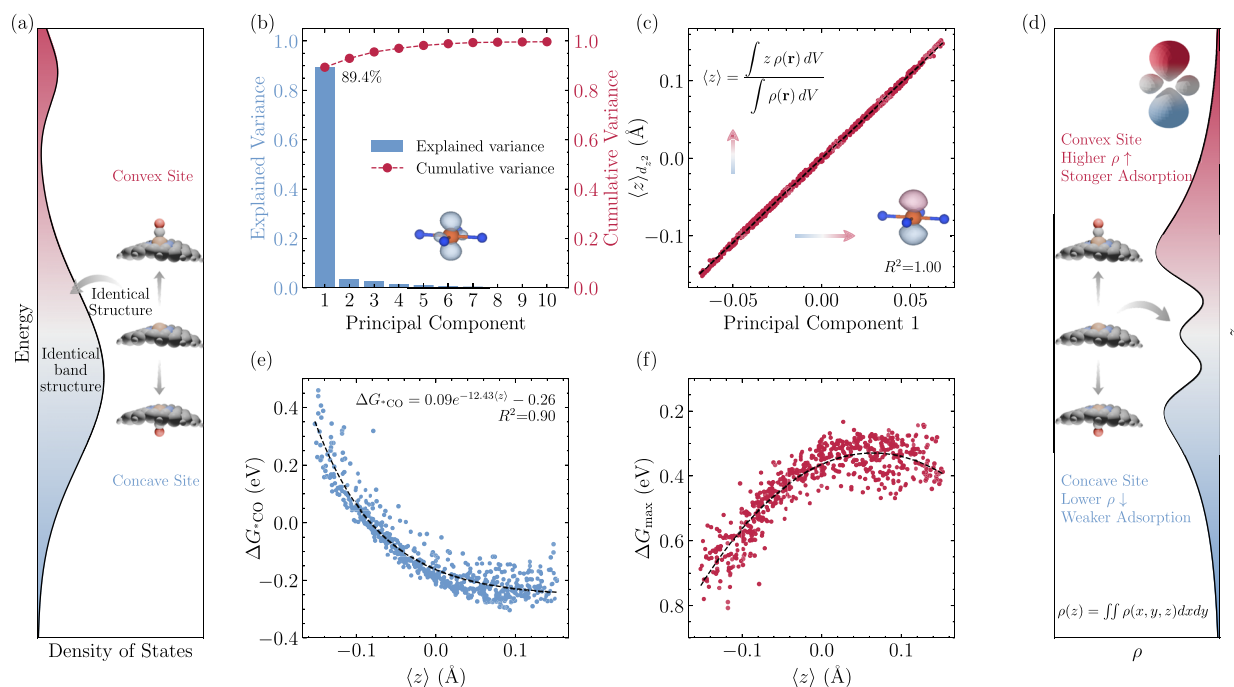


Figure 4. (a) Schematic demonstrating the inability of the d -orbital energy centroid model to distinguish between the catalytically nonequivalent upper (convex) and lower (concave) surfaces of a curved structure. (b) Variance of the d_z^2 orbital probability density explained by each principal component and cumulative variance. (c) Visualization of the PC1 loading vector of the d_z^2 orbital and its linear correlations with $\langle z \rangle_{d_z^2}$. (d) Schematic illustrating that curvature induces an asymmetric polarization of the d -orbital density (ρ) along the z -axis. Higher electron density accumulation on the adsorption side facilitates orbital overlap, leading to stronger binding, whereas density depletion results in weaker adsorption. (e) Plots of the adsorption free energies for ΔG_{*CO} as a function of the electronic descriptor $\langle z \rangle$. (f) Plot of catalytic activity ($\Delta G_{\max}$) against the electronic descriptor $\langle z \rangle$.

with the quantum-mechanical nature of orbital interactions. Because atomic wavefunctions decay exponentially in the classically forbidden region, the overlap between the metal d_z^2 orbital and the adsorbate is expected to decrease approximately exponentially with separation, leading to a comparable distance dependence of the interaction strength.⁶³ The same exponential scaling holds for $*CO_2$ and $*COOH$ (Figure S15), although the correlation for $*COOH$ is noticeably weaker. This observation now explains the differential sensitivity previously noted. CO_2 adsorption preserves the local Fe spin state ($\mu: 2\mu_B \rightarrow 2\mu_B$), indicating weak Fe– CO_2 coupling; however, the activated V-shaped CO_2 still engages the same orbital channels as CO (σ -donation into d_z^2 and back-donation from d_{xz}/d_{yz} into $CO_2\pi^*$), but with a smaller magnitude (Figure S13c–e). Because the two adsorbates interact with Fe through symmetry-equivalent orbital channels, their responses to $\langle z \rangle$ are proportional, yielding a strong mutual scaling relation. $COOH$, by contrast, binds through a single σ -radical pairing between its unpaired carbon radical and the singly occupied d_z^2 , as evidenced by partial spin quenching ($\mu: 2\mu_B \rightarrow 1\mu_B$). The absence of effective π -back-donation channels imposes an intrinsic saturation limit on the energetic response to $\langle z \rangle$, producing the weaker scaling. This mechanistic distinction implies that curvature modulates different intermediates through inequivalent electronic pathways, potentially enabling differential tuning beyond conventional linear scaling relations. Despite this adsorbate-specific modulation, the monotonic dependence on $\langle z \rangle$ persists across all intermediates, establishing

a direct link between local geometry and reactivity that ultimately yields a well-defined volcano-shaped relationship (Figure 4f).

3.5. Mechanical Origin of the Geometric Descriptor

With $\langle z \rangle$ established as the intrinsic electronic variable that bypasses the energetic convolution identified earlier, the remaining question is mechanistic: how does macroscopic curvature encode itself into this microscopic orbital polarization? To answer this, we decompose the orbital polarization response along each orthogonal curvature mode. Under both isotropic (K_i) and anisotropic (K_a) curvature modes, $\langle z \rangle$ exhibits a strictly odd-symmetric response (Figure 5a,b). As the curvature magnitude increases, the response deviates from linearity and approaches saturation, accurately captured by a hyperbolic tangent function,

$$\langle z \rangle_j = B_j \tanh(\kappa_j K_j) \quad (14)$$

where B_j represents the maximum attainable saturated displacement, and κ_j controls the rate at which polarization develops with increasing curvature. Quantitative fitting reveals a pronounced hierarchy between the two modes. The anisotropic component exhibits a substantially larger saturation amplitude ($B_a \approx 0.16$) compared to the isotropic mode ($B_i \approx 0.10$), consistent with the broader displacement range observed directly from the DFT data (0.30 vs. 0.18 Å). Moreover, the magnitude of $|\kappa_a|$ (~ 7.44) slightly exceeds that of $|\kappa_i|$ (~ 6.78), indicating a

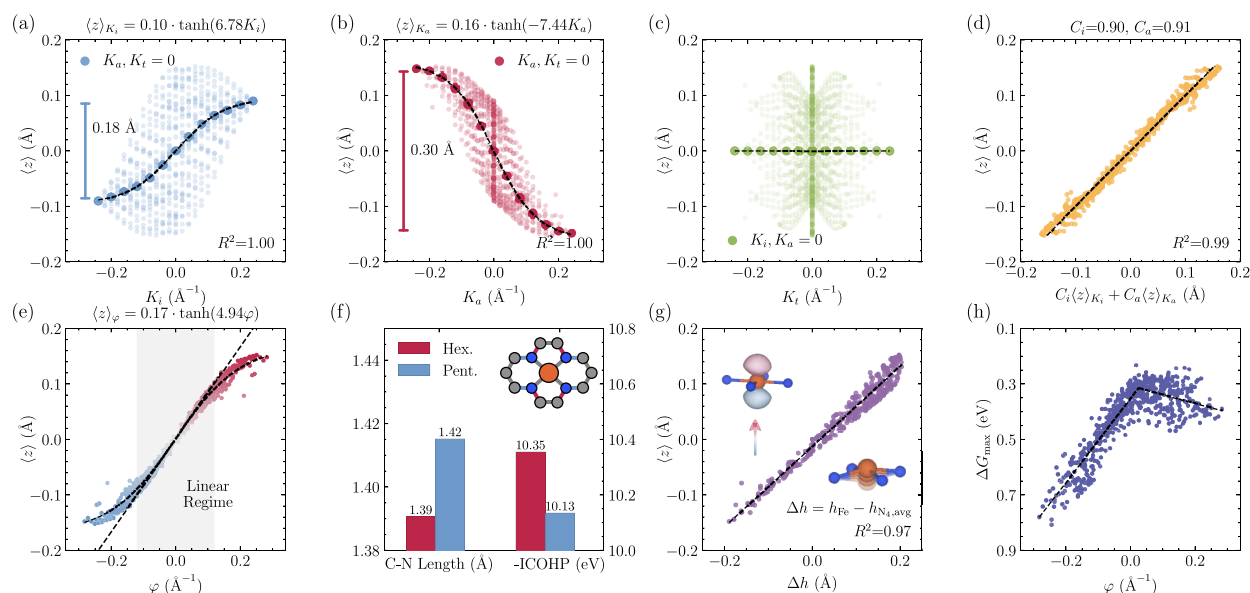


Figure 5. Correlation between $\langle z \rangle$ and (a) isotropic (symmetric) curvature parameter K_p , (b) anisotropic curvature parameter K_a , and (c) torsional (twist) curvature parameter K_t . In panels (a–c), transparent markers correspond to all sampled configurations, whereas large filled markers indicate the points obtained from controlled symmetry-constrained sampling. (d) Parity plot comparing $\langle z \rangle$ with values predicted by the multivariate linear regression model based on $\langle z \rangle_{K_i}$ and $\langle z \rangle_{K_a}$. (e) Plot of the electronic descriptor, $\langle z \rangle$, as a function of the geometric descriptor, φ . (f) Comparison of both bond lengths and -ICOHP values for the first and second coordination shells, analyzed along pentagonal and hexagonal ring directions across different coordination environments. (g) Structural origin of the electronic response: Linear dependence of $\langle z \rangle$ on the vertical Fe displacement (Δh). (h) Plot of the catalytic activity ($\Delta G_{\max}$) against the geometric descriptor, φ .

steeper initial polarization response under anisotropic bending. Together, these parameters demonstrate that K_a not only produces a larger ultimate orbital displacement but also drives a more rapid buildup of axial polarization, quantitatively rationalizing its dominant role in the energy hierarchy. The near-perfect agreement with calculations ($R^2 \approx 1.00$) establishes this hyperbolic tangent form as the intrinsic nonlinear law governing curvature-induced orbital polarization, reflecting the finite deformability of the d -electron cloud. In contrast, the torsional curvature mode K_t produces no discernible response in $\langle z \rangle$ (Figure 5c), with correlations close to zero. This observation is fully consistent with the symmetry-mandated vanishing of the odd term.

Consistent with the approximate mode additivity of ΔG_{SCO} ($R^2 = 0.96$), the underlying electronic variable $\langle z \rangle$ strictly follows a linear superposition model. Specifically, combining the isolated single-mode contributions,

$$\langle z \rangle \approx C_i \langle z \rangle_{K_i} + C_a \langle z \rangle_{K_a} \quad (15)$$

recovers the full orbital polarization landscape with near-perfect fidelity ($R^2 = 0.99$, Figure 5d), with fitted coefficients $C_i = 0.90$ and $C_a = 0.91$ both close to unity. While the nonlinear tanh-type model accurately captures the saturation behavior at extreme curvatures, the linear response regime ($K \rightarrow 0$) offers a simpler and more transferable description by retaining only the first-order susceptibilities and bypassing higher-order terms that are more sensitive to the global elastic boundary conditions of the support. Therefore, by performing a first-order Taylor

expansion ($\tanh(x) \approx x$), we express the polarization as a superposition of orthogonal geometric modes:

$$\begin{aligned} \varphi &\approx C_i \langle z \rangle_{K_i} + C_a \langle z \rangle_{K_a} \\ &\approx C_i B_i \tanh(\kappa_i K_i) + C_a B_a \tanh(\kappa_a K_a) \\ &\approx C_i B_i \kappa_i K_i + C_a B_a \kappa_a K_a \\ &= c_i K_i + c_a K_a \end{aligned} \quad (16)$$

where $c_j = C_j B_j \kappa_j$ denotes the intrinsic geometric susceptibility. To delineate the validity of this linear description, we systematically compare the linear prediction φ with the DFT-calculated orbital displacement $\langle z \rangle$ (Figure 5e). Within the curvature range $|K| < 0.11 \text{ \AA}^{-1}$, all data points cluster tightly around the parity line, with deviations remaining below 10%. Translating this threshold into geometric dimensions, this linear regime corresponds to structural features with radii of curvature $R \gtrsim 9 \text{ \AA}$, equivalent to CNT diameters of $D > 1.8 \text{ nm}$ or spherical pore diameters of $D > 3.6 \text{ nm}$. This range encompasses the vast majority of experimentally synthesized catalytic supports, including CNTs and the predominant fraction of mesoporous carbons, thereby confirming that the simple linear descriptor is sufficiently accurate for most practical catalyst design scenarios. At larger curvature (e.g., ultrasmall fullerenes or small-diameter CNTs), the true displacement systematically falls below the linear prediction, exhibiting a characteristic S-shaped deviation that signals the progressive involvement of higher-order nonlinear terms.

Substituting the fitted parameters yields $c_i \approx 0.58$ and $c_a \approx -1.07$ Å. The opposing signs indicate that the two curvature modes drive orbital polarization in opposite directions, while the nearly twofold disparity in magnitude ($|c_a| \approx 2|c_i|$) confirms the dominance of anisotropic distortion even in the perturbative limit. To decode the microscopic origin of this sign contrast, we project φ back onto the local bond axes (k_x, k_y) using the inverse relations:

$$\begin{aligned}\varphi &= (c_i + c_a) k_x + (c_i - c_a) k_y \\ &\equiv c_x k_x + c_y k_y \\ &\approx -0.48 k_x + 1.65 k_y\end{aligned}\quad (17)$$

This decomposition reveals a striking competition: curvature along the pentagonal-ring direction (k_y) has a dominant positive coefficient ($c_y \approx 1.65$), whereas curvature along the hexagonal-ring direction (k_x) has a smaller, negative coefficient ($c_x \approx -0.48$). This implies that positive bending along the y -axis efficiently drives the electron cloud upward (increasing $\langle z \rangle$), while the same positive bending along the x -axis paradoxically pulls it downward (reducing $\langle z \rangle$), albeit less effectively. We trace this direction-dependent behavior to the contrasting mechanical stiffness of the coordination environment. Our calculations confirm that C–N bonds in the pentagonal rings are shorter (1.39 vs. 1.42 Å) and intrinsically stronger (ICOHP: 10.35 vs. 10.13 eV) than those in the hexagonal rings (Figure S5f). Consequently, the rigid pentagonal rings resist in-plane deformation, channeling geometric strain into a pronounced out-of-plane protrusion of the metal center; in contrast, the more flexible hexagonal rings partially dissipate strain through in-plane lattice relaxation, resulting in a dampened, negative displacement. This mechanical displacement (Δh) serves as the direct structural proxy for the electronic polarization, evidenced by the precise linear scaling between $\langle z \rangle$ and Δh (Figure S5g). This mechanical picture also resolves the puzzling inertness of the twist mode K_t noted earlier. Since $K_t = 2k_{xy}$ is the off-diagonal component of the curvature tensor, a pure K_t deformation (with $K_i = K_a = 0$) implies $k_x = k_y = 0$, meaning zero net bending along either bond-axis direction. Without differential strain along the pentagonal and hexagonal rings, the mechanical driving force for vertical Fe displacement vanishes, rendering K_t incapable of producing a significant $\langle z \rangle$ response. Armed with this physically grounded descriptor, we finally resolve the geometric complexity: plotting the catalytic activity ($\Delta G_{\max}$) against φ consolidates the entire 602-point landscape onto a single, pristine volcano (Figure 5h), validating φ as the unified predictive coordinate for curvature-driven catalysis.

3.6. Generality and Design Implications of the Geometric Descriptor

We next assess whether φ retains its predictive power beyond the Fe–N₄–C reference system. Validation across three independent dimensions—chemical identity (M = Mn, Co, Ni; adsorbates *OH, *H, *N), first-shell coordination topology (N₃C, N₃V, and N₂C₂ isomers), and computational framework (vacuum vs. –0.8 V vs. SHE; constrained clusters vs. periodic supercells)—consistently preserves the monotonic φ –adsorption energy scaling (Figures 6 and S16; see Note S7 for full computational details), with chemical identity modulating only

slope and intercept while leaving the geometric driving force intact. This transferability is structurally rooted in the differential mechanical stiffness of bonds to five- vs. six-membered rings, a feature that survives compositional substitution in the first coordination shell (Figure S17; see Note S7 for ICOHP analysis). The co-monotonic response of *H also constrains the curvature dependence of the competing HER side reaction: although curvature simultaneously modulates HER and CO₂RR through the same out-of-plane d -orbital channel, the relative difference $\Delta G_{*H} - \Delta G_{*COOH}$ remains nearly constant across the entire curvature range (Figure S18), indicating that the CO₂RR-versus-HER selectivity window is preserved under curvature engineering.

The generality of φ further manifests in its ability to subsume earlier, more restricted descriptors as limiting cases. The linear curvature–adsorption correlations established for CNTs (uniaxial limit, $k_2 = 0$)^{36–40} and spherical carbon supports (isotropic limit, $k_1 = k_2$)²³ correspond to restricted slices of the complete geometric parameter space naturally recovered by φ under the respective high-symmetry constraints. Crucially, only by retaining the full curvature tensor does one resolve the apparent contradiction between studies reporting that armchair-direction bending weakens adsorption^{37–39} and those finding that zigzag-direction bending strengthens it:⁴⁰ these opposing observations correspond precisely to the opposite signs of $c_x < 0$ and $c_y > 0$, identifying anisotropic mechanical response as the hidden variable underlying both trends.

These insights translate directly into design principles that go beyond current practice. For CNT-supported catalysts, our analysis demonstrates that specifying the nanotube diameter is necessary but not sufficient: the chirality-dependent orientation (θ) of the Fe–N₄ axis relative to the local curvature vector produces a wide distribution of intrinsic activities even among tubes of identical diameter (Figure S19), implying that chirality control must be incorporated as an explicit design variable. More broadly, we note that planar M–N–C models frequently predict weak or even unfavorable initial CO₂ activation, yet many experimentally realized M–N–C catalysts—which are typically supported on curved carbon frameworks—exhibit appreciable CO₂RR activity. While multiple factors likely contribute to this discrepancy, the curvature-induced enhancement of CO₂ activation identified here may be one of them. Achieving deterministic rather than statistical exploitation of this effect—for instance, through biaxially strained arrays⁶⁴ or template-assisted curvature engineering—constitutes a tractable and high-impact direction for future catalyst design.

4. CONCLUSIONS

In summary, we have established a geometric framework grounded in differential geometry and a symmetry-constrained Taylor expansion to quantify the impact of surface curvature on single-atom catalysis. By decomposing the local topology of any smooth surface into orthogonal curvature modes— isotropic (K_i), anisotropic (K_a), and torsional (K_t)—and applying symmetry selection rules together with a small-curvature linear approximation, we derived the scalar geometric descriptor $\varphi = c_i K_i + c_a K_a$. Instantiating this formalism for Fe–N₄–C SACs, we demonstrate that φ maps the multidimensional CO₂RR landscape of 602 nonequivalent curved sites into a

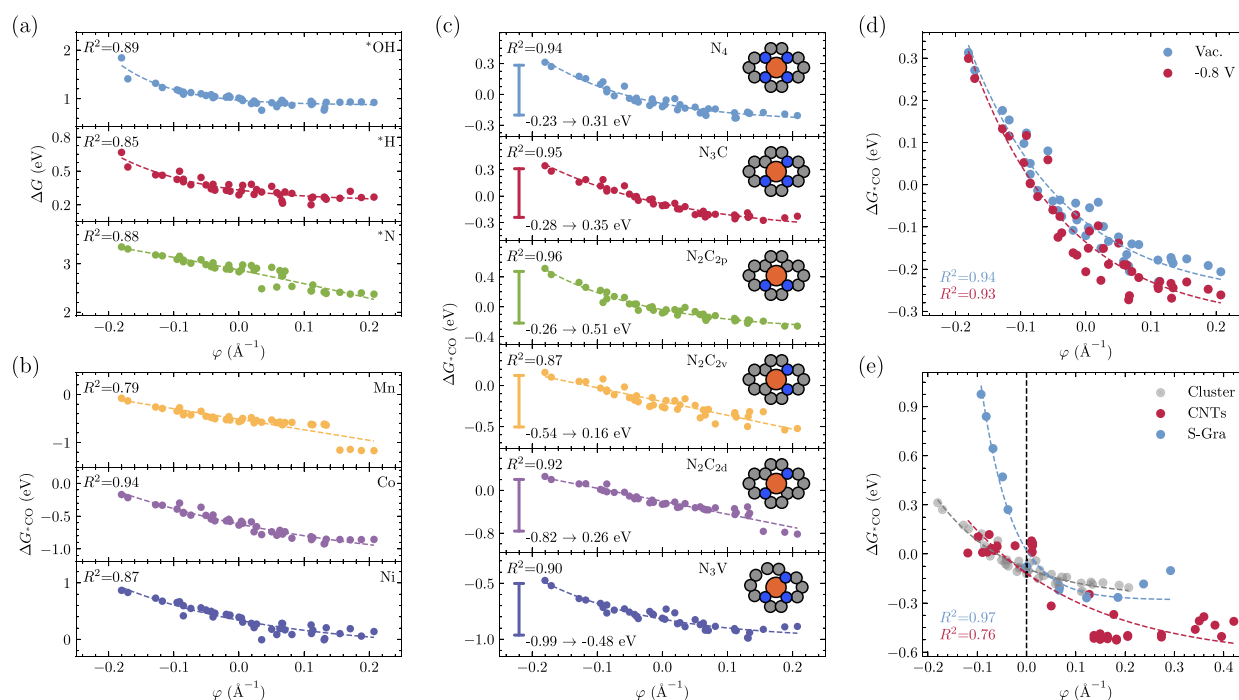


Figure 6. Transferability of the geometric descriptor ϕ across different chemical identities, local coordination environments, and modeling conditions. (a) Adsorption free energies of diverse intermediates (*OH, *H, and *N) on Fe-N₄-C as a function of ϕ . (b) Adsorption free energies of *CO on M-N₄-C (M = Mn, Co, and Ni) as a function of ϕ . (c) Adsorption free energies on Fe sites with distinct first-shell coordination environments (N₄, N₃C, N₂C_{2p}, N₂C_{2v}, N₂C_{2d}, and N₃V) as a function of ϕ . (d) ΔG_{*CO} on Fe-N₄-C computed under vacuum and under an applied electrochemical potential of -0.8 V vs. SHE as a function of ϕ . (e) ΔG_{*CO} as a function of ϕ for uniaxially strained graphene, carbon nanotubes of varying diameters and chiral indices, and their overlay with the cluster-library data.

predictive volcano relationship. The underlying electronic origin is an asymmetric out-of-plane *d*-orbital polarization along the surface normal, quantified by the out-of-plane centroid of the *d* orbital, $\langle z \rangle$. We trace this polarization to its mechanical origin: the intrinsic stiffness contrast between pentagonal and hexagonal rings channels geometric strain into opposite vertical displacements of the metal center. This establishes a causal pathway from geometry (ϕ) to mechanics, electronics ($\langle z \rangle$), and activity ($\Delta G_{\max}$).

Robustness is demonstrated across diverse metal centers, adsorbates, coordination environments, periodic models, and electrochemical conditions; recovering uniaxial and isotropic models as limiting cases further rationalizes the reported directional inconsistencies in the literature, identifying anisotropic mechanical response as the key governing variable. The persistence of a monotonic ϕ -activity response across all tested scenarios supports curvature engineering as a practical design axis complementary to conventional electronic-structure tuning. We anticipate that continued advances in the synthetic control of well-defined curvature fields will transform curvature from a statistical consequence of morphology into a controllable parameter in catalyst design.

■ ASSOCIATED CONTENT

Data Availability Statement

Code for structure generation, wavefunction analysis, and calculation of the electronic descriptor $\langle z \rangle$ is available at <https://github.com/Migie-cphy/Curvature-Driven-Catalysis>. The dataset, including DFT input/output files and exported orbital

densities, is archived on Zenodo (DOI: 10.5281/zenodo.18823122).

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.6c02847>.

Theoretical derivations, curved-model construction details, molecular dynamics protocols, symmetry-based dataset reduction, PCA analysis of *d*-orbital wavefunctions, generality and robustness tests, quadratic-fitting validation, and projected density of states (PDOS) analysis (PDF)

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Notes

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